

12 C

Gravitational field strength $g = \frac{GM}{r^2}$ (vector)

Due to different masses of the Earth and Moon, the field strength due to the Earth (points to the left) is larger than the field strength due to the Moon (points to the right). Hence the resultant field strength is not zero and points towards the Earth. A and B are wrong.

Gravitational potential $\phi = -\frac{GM}{r}$ (scalar)

Due to different masses of the Earth and Moon, the potential does not add to zero at point P. That would eliminate D.

13 D